Vivek Balasaheb Sabale

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RESEARCH INTERESTS

Quantum Information and Computation, Open Quantum Systems, Quantum Algorithms, Non-Markovian Dynamics.

EDUCATION

Indian Institute of Technology, Jodhpur, India 2021 – Present
Ph.D. in Quantum Information and Computation
Thesis Title: Correlation Structures in Multi-Qubit Quantum Systems, Quantum Channels, and Open Quantum Systems Dynamics

This thesis establishes a unified framework for characterizing, protecting, and modeling quantum correlations across multi-qubit architectures, correlated channels, and open-system dynamics. It introduces the Cumulant-Norm Entanglement Index (CNEI) to diagnose genuine multipartite correlations, employs weak measurements to mitigate environmental noise and enhance universal quantum teleportation, and examines memory effects in correlated non-Markovian channels.

Indian Institute of Technology, Madras, India 2019 – 2021
M.Sc. in Chemistry
Thesis Title: Entangled quantum states and correlations in quantum mechanics.

The thesis is exploration of multi-qubit entangled quantum states and how to identify the entanglement. It presents the use of Cumulants or correlation coefficients to diagnose genuine pure multipartite entanglement.

Fergusson College, Pune, India 2016 – 2019
B.Sc. in Chemistry

TEACHING EXPERIENCE

Teaching Assistant (TA), Indian Institute of Technology, Jodhpur

- **Chemistry Laboratory Courses:** Assisted in conducting undergraduate laboratory experiments, managing experimental setups, guiding students on safety protocols, and evaluating laboratory reports.
- **Quantum Information & Foundations:** Conducted tutorial sessions, assisted in problem sets preparation, and mentored students on core conceptual topics including quantum entanglement, and quantum channels.
- **Computational Hands-on Workshops (MATLAB & Mathematica):** Led practical sessions demonstrating numerical modeling, data handling, and symbolic computation techniques in MATLAB and Mathematica for physical chemistry students.

RESEARCH PUBLICATIONS

Journal Articles

1. S. Bhattacharya, **V. B. Sabale**, and A. Kumar, "Heisenberg spin networks for realizing quantum battery with the aid of Dzyaloshinskii-Moriya interaction," *New Journal of Physics*, vol. 28, no. 1, p. 014508, 2026. DOI: 10.1088/1367-2630/ae33e4

2. D. Sharma, **V. B. Sabale**, P. Singh, and A. Kumar, "Harnessing quantum support vector machines for cross-domain classification of quantum states," *Quantum Machine Intelligence*, vol. 7, no. 1, p. 49, 2025. DOI: 10.1007/s42484-025-00274-4

3. A. Abu-Nada, S. Banerjee, and **V. B. Sabale**, "Exploring the non-markovian dynamics in depolarizing maps," *Phys. Rev. A*, vol. 110, p. 052209, Nov 2024. DOI: 10.1103/PhysRevA.110.052209

4. **V. B. Sabale**, N. R. Dash, A. Kumar, and S. Banerjee, "Facets of correlated non-markovian channels," *Annalen der Physik*, vol. 536, no. 10, p. 2400151, 2024. DOI: 10.1002/andp.202400151

5. **V. B. Sabale**, A. Kumar, and S. Banerjee, "Toward realization of universal quantum teleportation using weak measurements," *Annalen der Physik*, vol. 536, p. 2300392, Apr 2024. DOI: 10.1002/andp.202300392

CONFERENCES & PRESENTATIONS

Quantum Techniques in Machine Learning (QTML 2024) Nov 2024
University of Melbourne, Australia

- Poster presentation about article *"Harnessing Quantum Support Vector Machines and Neural Networks for Cross-Domain Classification of Multi-Qubit Quantum States"*.

Quantum Information Processing (QIP 2024) Jan 2024
Taipei, Taiwan

- Poster presentation about article *"Toward realization of universal quantum teleportation using weak measurements"*.

Theoretical Chemistry Symposium (TCS 2023) 2023
Indian Institute of Technology, Madras

- Poster presentation about article *"Facets of correlated non-markovian channels"*.

AWARDS & ACHIEVEMENTS

International Travel Grant, Anusandhan National Research Foundation (ANRF) 2024

Received full financial travel assistance from the Government of India (ANRF) to participate in the international conference *Quantum Techniques in Machine Learning (QTML 2024)* at the University of Melbourne, Australia (Nov 25 – Nov 29, 2024). Presented poster on ongoing research on leveraging quantum kernel methods, Quantum Support Vector Machines (QSVM), and quantum neural networks for classification of quantum states.

Mahatma Jyotirav Phule Gunvant Vidyarthi Scholarship, Pune University 2018

CERTIFICATIONS & TRAINING

IBM Quantum Challenge Certificate, IBM 2021
Awarded for completing advanced hands-on challenges in quantum computing using Qiskit. Solved complex computational problems in quantum algorithm design, quantum gate optimization, and circuit implementation on real quantum hardware.

Diploma Certificate, Project Oriented Chemistry Education (POCE), JNCASR, Bangalore (2016–2019)
A prestigious three-year competitive summer research fellowship program by Jawaharlal Nehru Centre for Advanced Scientific Research (JNCASR). Participated in intensive summer research projects, advanced chemistry lectures, and laboratory training under expert mentors.

MISCELLANEOUS EXPERIENCE

Workshop Resource / Instructor, QIndia & IIIT Dharwad 2024
Taught quantum computing during an interactive workshop organized at IIIT Dharwad.

SKILLS

- **Coding & Frameworks:** MATLAB, Mathematica, Python, Qiskit, Julia
- **Languages:** English, Hindi, Marathi (Strong reading, writing, and speaking competencies)

REFERENCES

Prof. Atul Kumar
Professor
Indian Institute of Technology, Jodhpur
Website